DPU

Unit-V
Machine Translation (MT)

Syllabus Outline

- Need of MT, Problems of Machine Translation,
- MT Approaches, Direct Machine Translations,
- Rule Based Machine Translation,
- Knowledge Based MT System,
- Statistical Machine Translation (SMT),
- Parameter learning in SMT (IBM models) using EM),
- Encoder-decoder architecture,
- Neural Machine Translation

Machine Translation (MT)

- **Machine Translation** (MT) is a domain of computational linguistics that uses computer programs to translate text or speech from one language to another with no human involvement with the goal of relatively high accuracy, low errors, and effective cost.
- Machine Translation is a very important yet complex process as there is currently a huge number of nuanced natural languages in the world.

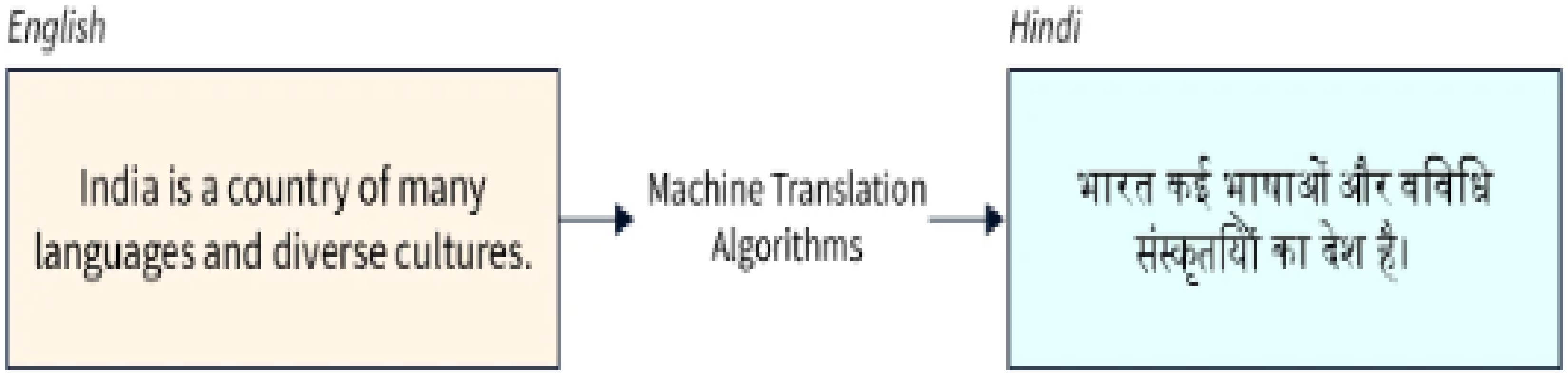
Need for Machine Translation

- There is an explosion of data in the modern world with the information revolution and since language is the most effective medium of communication for humans, there is also an increased need for translating from one language to another using NLP tools and other techniques.

- The demand for translation is majorly due to the exponential rise increase in the exchange of information between various regions using different regional languages.

- **Examples** such as access to web documents in non-native languages, using products from across countries, real-time chat, legal literature, etc. are some use cases.

Need for Machine Translation



Why is machine translation so hard to tackle?

Machine translation using NLP tools from one language to another is notoriously difficult due to the presence of many words with multiple meanings, sentences with potential possible readings, and the fact that certain grammatical relations in one language might not exist in another language.

What are the benefits of machine translation?

Fast Translation Speed:

- Machine translation using NLP has the potential to translate millions of words for high-volume translations happening in modern day-to-day processes.

- MT cuts the human translator effort usually done post-editing.

- We can use machine translation to tag high-volume content and organize the contents so that we can look up over search or retrieve content in translated languages quickly.

What are the benefits of machine translation?

Excellent Language Selection:

- Most machine language translation tools these days can translate more than 100 languages.
- The MT programs are powerful enough to translate multiple languages at once so they can be rolled out to a global user base across products and do documentation updates on the fly.
- MT is well-suited to many language pairs these days (English to most European languages) with very high accuracies.

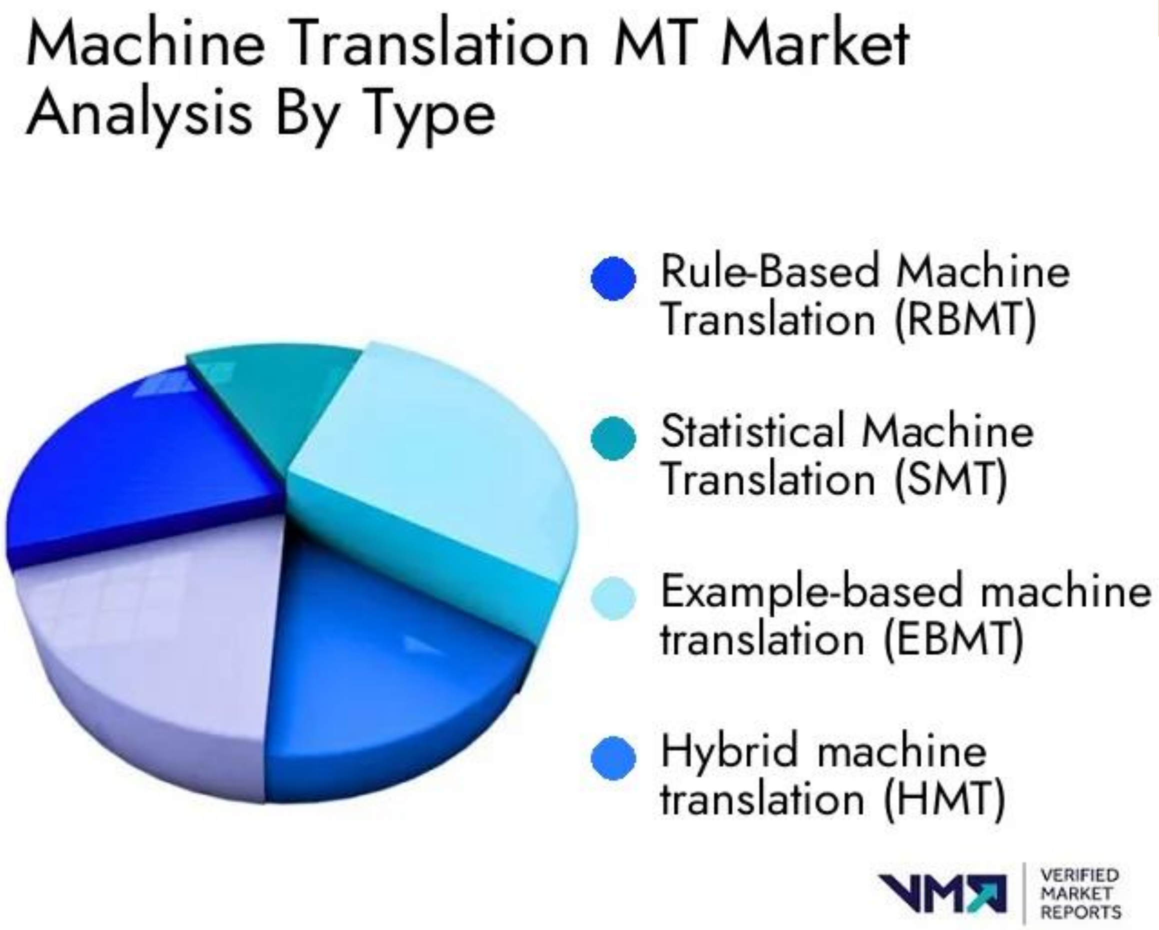
Reduced Costs:

- As annotation with humans doesn't scale well both in terms of cost and speed, we can use machine translation using NLP to cut translation delivery times and costs.

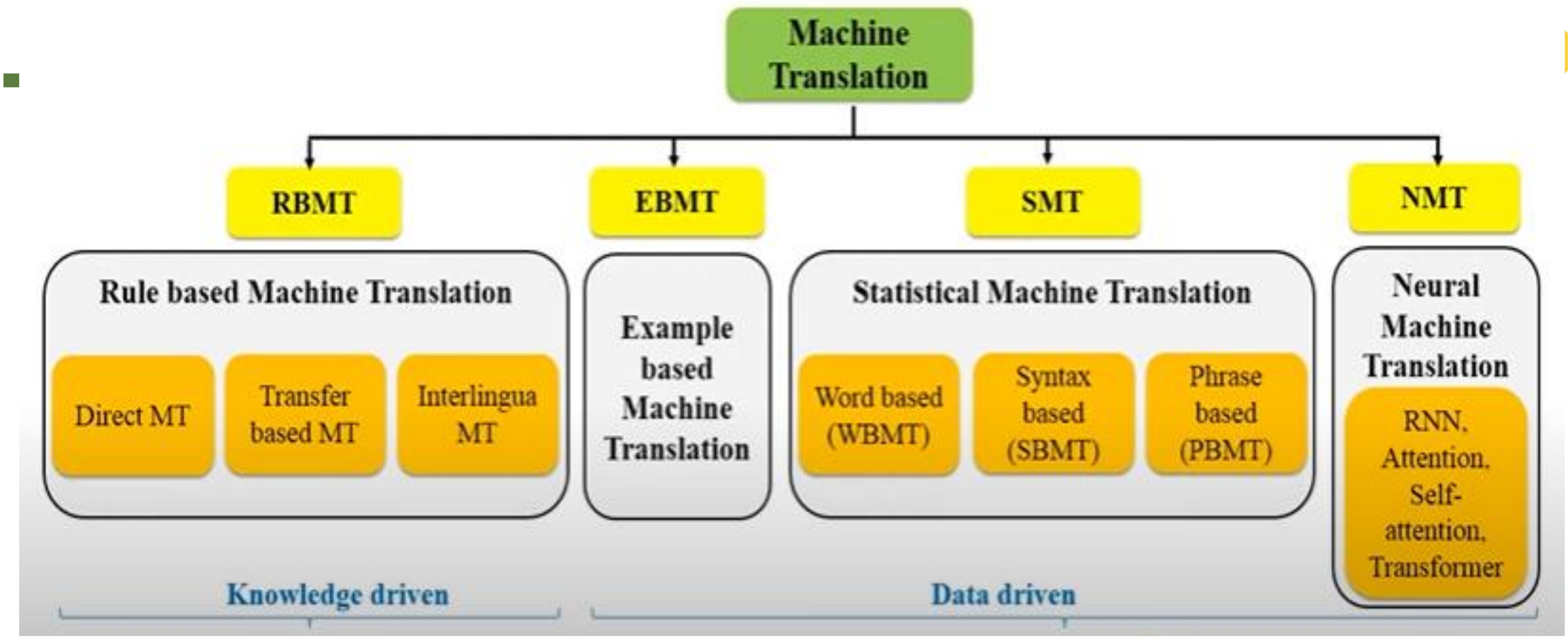
We can also produce basic translations with little compute costs that can be further

MT Approaches

- Rule-based Machine Translation or RBMT
- Hybrid Machine Translation or HMT
- Neural Machine Translation or NMT
- Statistical Machine Translation or SMT



Approaches to MT



- **Rule-based Machine Translation or RBMT**

- RBMT basically translates the basics of *grammatical rules*.
- It directs a grammatical examination of the source language and the objective language to create the translated sentence.
- But, RBMT requires broad editing, and its substantial reliance on dictionaries implies that proficiency is accomplished after a significant period.

Rule-based Machine Translation or RBMT

- It is also called **knowledge-based machine translation**, these are the earliest set of classical methods used for machine translation.
- These translation systems are mainly based on **linguistic information** about the source and target languages that are derived from dictionaries and grammar covering the characteristic elements of each language separately.

Approaches to MT

RBMT

- Components:
  - **Bilingual dictionary (German-English)**
  - **Set of linguistic rules**
  - **Other tools such spell checker etc.**

RBMT

Rule based Machine Translation

Direct MT	Transfer based MT	Interlingua MT

Approaches to MT

RBMT

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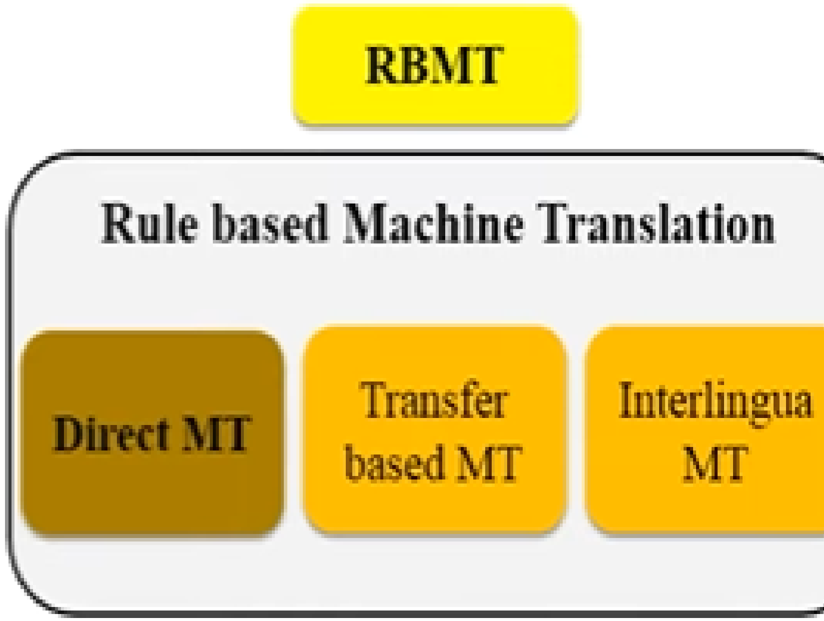
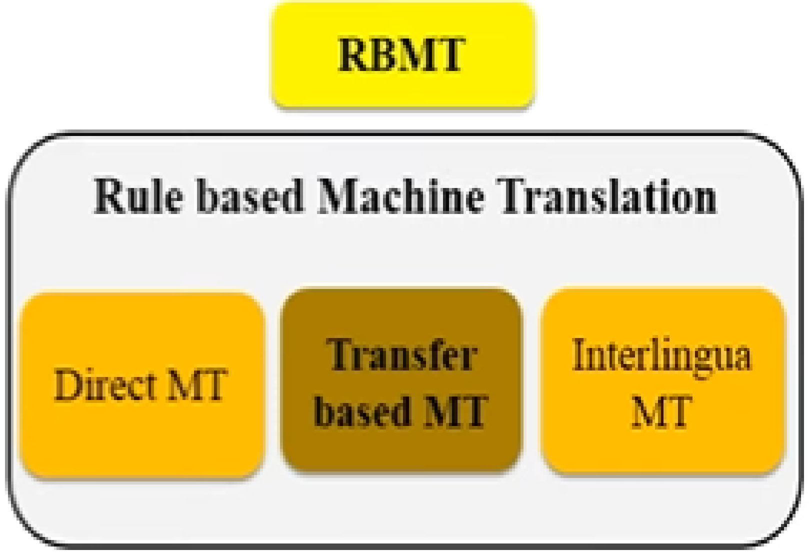


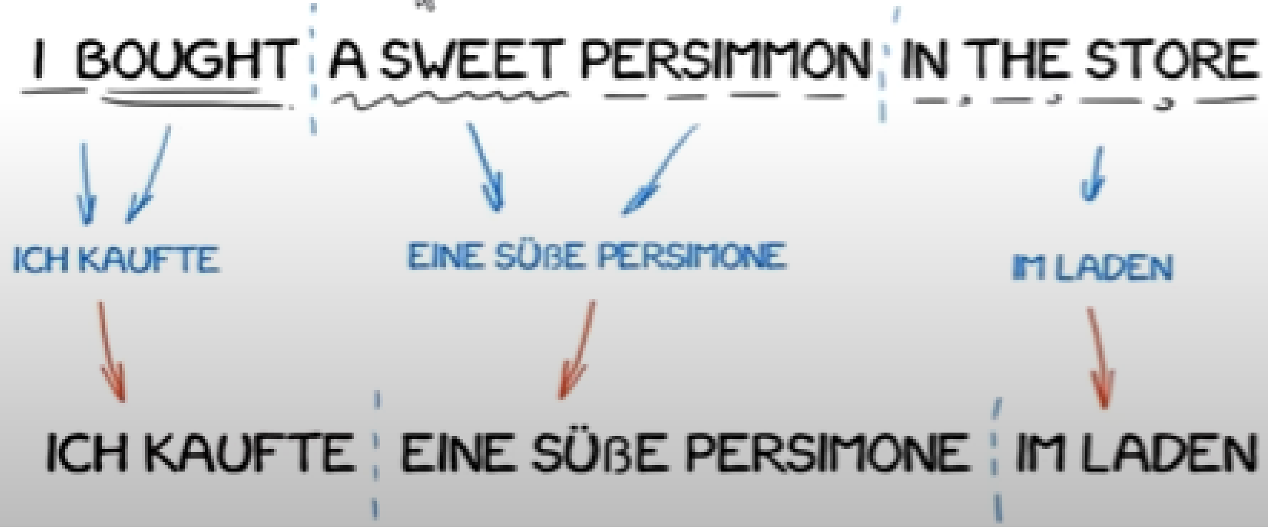
Approaches to MT

RBMT

- Components:
  - **Bilingual dictionary (German-English)**
  - **Set of linguistic rules**
  - **Other tools such spell checker etc.**



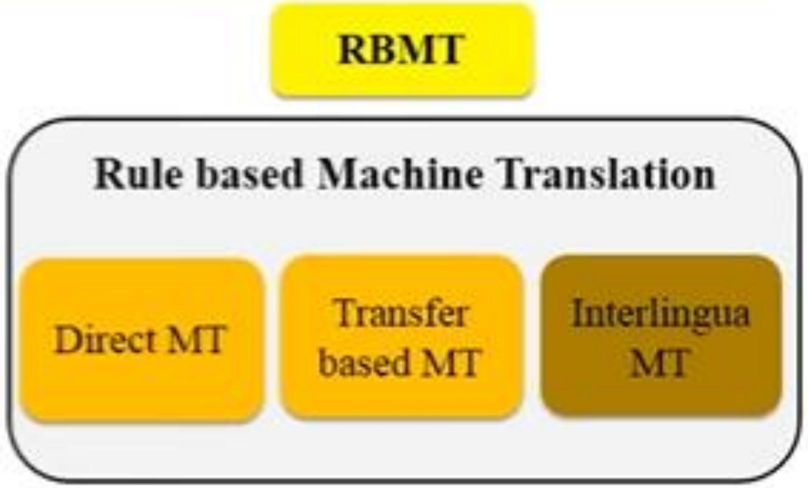
- Understand the grammatical structure of the source sentence then translate



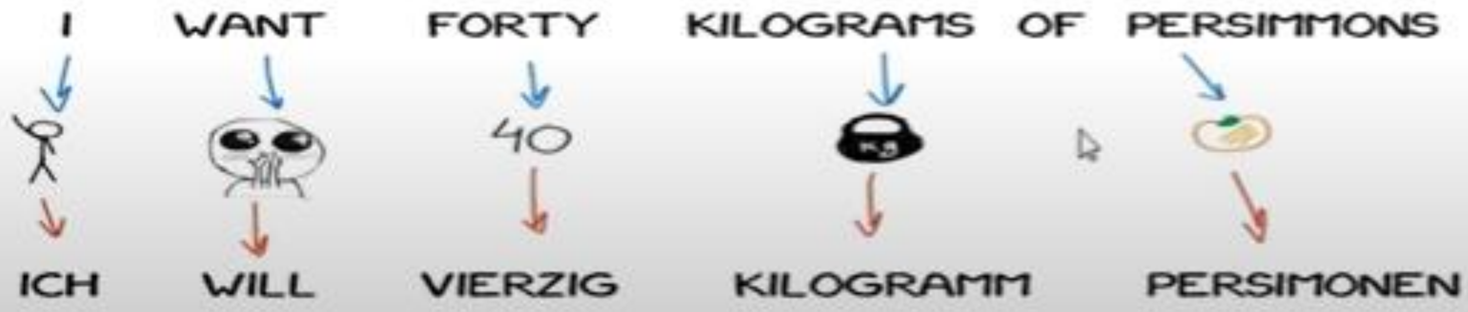
Approaches to MT

RBMT

- Components:
  - **Bilingual dictionary (German-English)**
  - **Set of linguistic rules**
  - **Other tools such spell checker etc.**



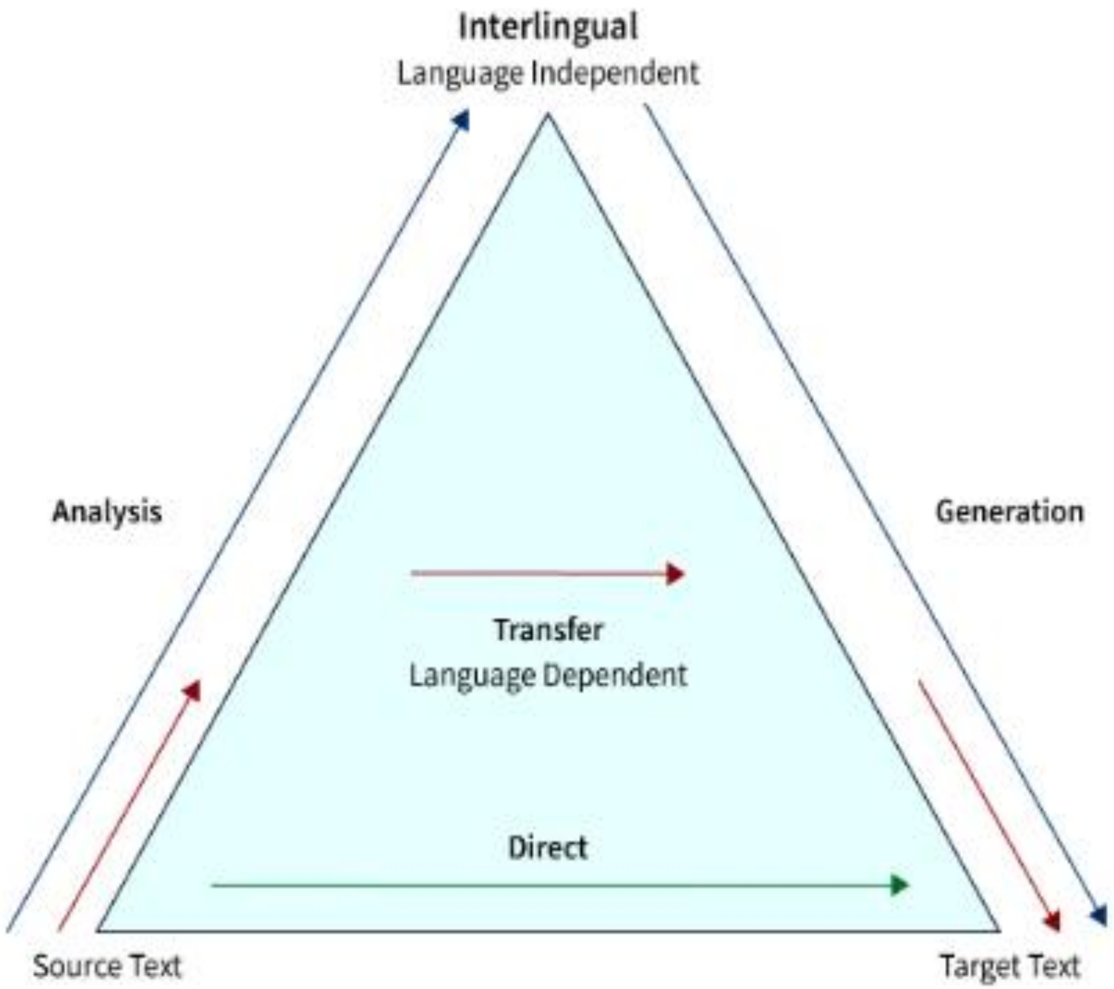
- First represent meaning of source sentence into language independent representation then translate



Rule-based Machine Translation or RBMT

- Once we have input sentences in some source languages, RBMT systems generally generate the translation to output language based on the morphological, syntactic, and semantic analysis of both the source and the target languages involved in the translation tasks.
- The sub-approaches under RBMT systems are the direct machine translation approach, the interlingual machine translation approach, and the transfer-based machine translation approach.

- **<u>Types of Machine Translation</u>**

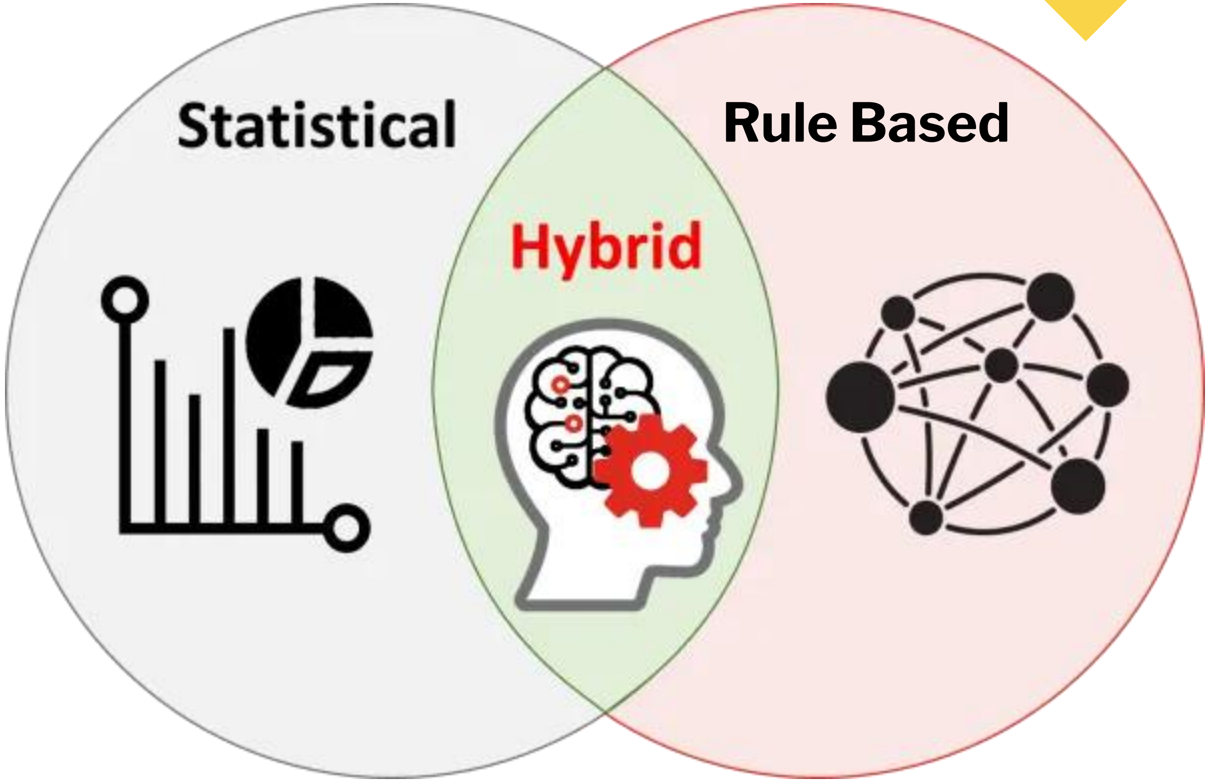


Hybrid Machine Translation or HMT

- HMT, as the term demonstrates, is a mix of RBMT and SMT.
- It uses a translation memory, making it unquestionably more successful regarding quality.
- Nevertheless, even HMT has a lot of downsides, the biggest of which is the requirement for enormous editing, and human translators will also be needed.
- There are several approaches to HMT like multi-engine, statistical rule generation, multi-pass, and confidence-based.

- **Hybrid Machine Translation or HMT**

- Hybrid Machine Translation approaches take advantage of both statistical and rule-based translation methodologies which were proven to have better efficiency in the area of machine translation systems using NLP.



• Hybrid Machine Translation or HMT

- These techniques are better than the previous methods and have more power, flexibility, and control in translation tasks in general.

- Most governmental and industrial-based machine translation systems use the hybrid-based approach to develop translation from source to target language, which is based on both rules and statistics.

- **Hybrid Machine Translation or HMT**

- Hybrid-based approaches combine the best of both worlds such that they can be used in several different ways.

- In cases where translations are performed in the first stage using a rule-based approach, they can be followed by adjusting or correcting the output using statistical information.

- On the other way, rules are used to pre-process the input data as well as post-process the statistical output of a statistical-based translation system.

Neural Machine Translation or NMT

- NMT is a type of machine translation that relies upon neural network models (based on the human brain) to build statistical models with the end goal of translation.
- The essential advantage of NMT is that it gives a solitary system that can be prepared to unravel the source and target text.
- Subsequently, it doesn't rely upon specific systems that are regular to other machine translation systems, particularly SMT.

- # Neural Machine Translation or NMT

- **Neural Machine Translation** methods in NLP employ artificial intelligence techniques and, in particular, rely upon neural network models which were originally based on the human brain to build machine learning models with the end goal of translation and also further learning languages, and improving the performance of the machine translation systems constantly.

• Neural Machine Translation or NMT

- Neural Machine Translation systems do not need any specific requirements that are regular to other machine translation systems like statistical-based methods in general.

- Neural Machine Translation works by incorporating training data, which can be generic or custom depending on the user’s needs.

➢ **Generic Data:** Includes the total of all the data learned from translations performed over time by the machine translation engines employed in the entire realm of translation.

This data also serves as a generalized translation tool for various applications,including text, voice, and documents.

Neural Machine Translation or NMT

- **Custom or Specialized Data:** These are the sets of training data fed to machine translation tasks to build specialization in a subject matter.

- Subjects include engineering, design, programming, or any discipline with its specialized glossaries and dictionaries.

- https://www.youtube.com/watch?v=NZTVm_0UTpc

- **Neural Machine Translation or NMT**

Advantages of NMT Models:

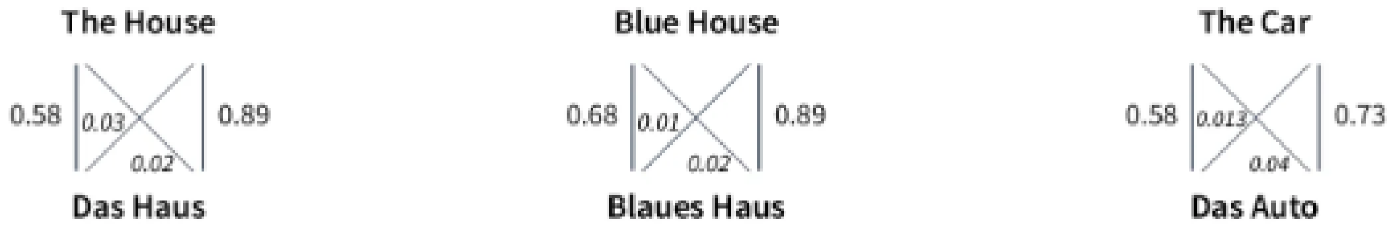
- NMT has achieved the state of the art performance in large-scale translation tasks in many languages like English to French, English to German etc.
- NMT requires minimal domain knowledge and is conceptually very simple and powerful.
- NMT has a small memory footprint as it does not store gigantic phase tables and language models.
- NMT also can generalize well to very long word sentences.

Statistical Machine Translation or SMT

- Statistical machine translation (SMT) models are generated with the use of statistical models whose parameters are derived from the analysis of bilingual text corpora.
- The ideas for SMT come from information theory.
- The principle theory of SMT-based models is that they rely on the Bayes Theorem to perform the analysis of parallel texts.
- The central idea is that every sentence in one language is a possible translation of any sentence in the other language, but the most appropriate is the translation that is assigned the highest probability by the system.

Statistical Machine Translation or SMT

- The SMT methods need millions of sentences in both the source and target languages for analysis to collect the relevant statistics for each word.
- In the early set of models, the abstracts of the European Parliament and the `United Nations Security Council meetings, which were available in the languages of all member countries, were taken for getting the required probabilities.
- SMT-based methods are further classified into Word-based SMT, Phrase-based SMT, and Syntax based SMT approaches.



Statistical Machine Translation or SMT

- *Google Translate*

Statistical Machine Translation or SMT

- Instead of relying on linguistic rules, statistical machine translation uses machine learning to translate text.
- The machine learning algorithms analyze large amounts of human translations that already exist and look for statistical patterns.
- The software then makes an intelligent guess when asked to translate a new source text.
- It makes predictions on the basis of the statistical likelihood that a specific word or phrase will be with another word or phrase in the target language.

Approaches to MT

SMT

Source (S) ⟶ Target (T)

$$\hat{T} = \underset{T}{argmax}\, P(T|S)$$

$$\hat{T} = \underset{T}{argmax}\ \underbrace{P(S|T)}_{\text{Translation model}}\ \underbrace{P(T)}_{\text{Language model}}$$

SMT

Statistical Machine Translation

- Word based (WBMT)
- Syntax based (SBMT)
- Phrase based (PBMT)

Bayes' rule

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

- **Translation model**: How probable T is as a translation of sentence S. What is the probability that S comes from T?
- **Language model**: How probable the sentence T is in target language. (helps in choosing the grammatically correct sentence. i.e. it will prefer 'I go' over 'go I')

Issues with SMT-based Approaches

- **Sentence Alignment**: As we are using two corpora to match sentences, many times we will encounter single sentences in the source language, which can be found translated into several sentences in the target language.
  - **Example:** When translating from English to German, the sentence "Raghu does not live here," the word "does" doesn't have a clear alignment in the translated sentence "Raghu wohnt hier nicht."
- **Statistical Anomalies**: Real-world training sets may override translations like proper nouns.
  - **Example:** "I took the train to Berlin" gets mistranslated as "I took the train to Paris" due to an abundance of "train to Paris" in the training set generally.
- **Idioms**: Depending on the corpora used, idioms may not translate idiomatically into the target language.
  - **, For example,**, using Canadian Hansard as the bilingual corpus, "hear" may almost invariably be translated to "Bravo!" since in Parliament, "Hear, Hear!" becomes "Bravo!".

Issues with SMT-based Approaches

- **Different Word Orders**: Since the word orders in languages differ, Statistical machine translation methods do not work well between languages that have significantly different word orders, **for example** Japanese and European languages.

- **Corpus Creation**: This can be very costly for users with limited resources, and sometimes the results are also unexpected where the superficial fluency can be deceiving.

- **Data Dilution**: This is one of the most common anomalies caused when attempting to construct a new statistical model to represent a distinct terminology for a specific corporate brand or domain.

Problem at hand-Machine Translation

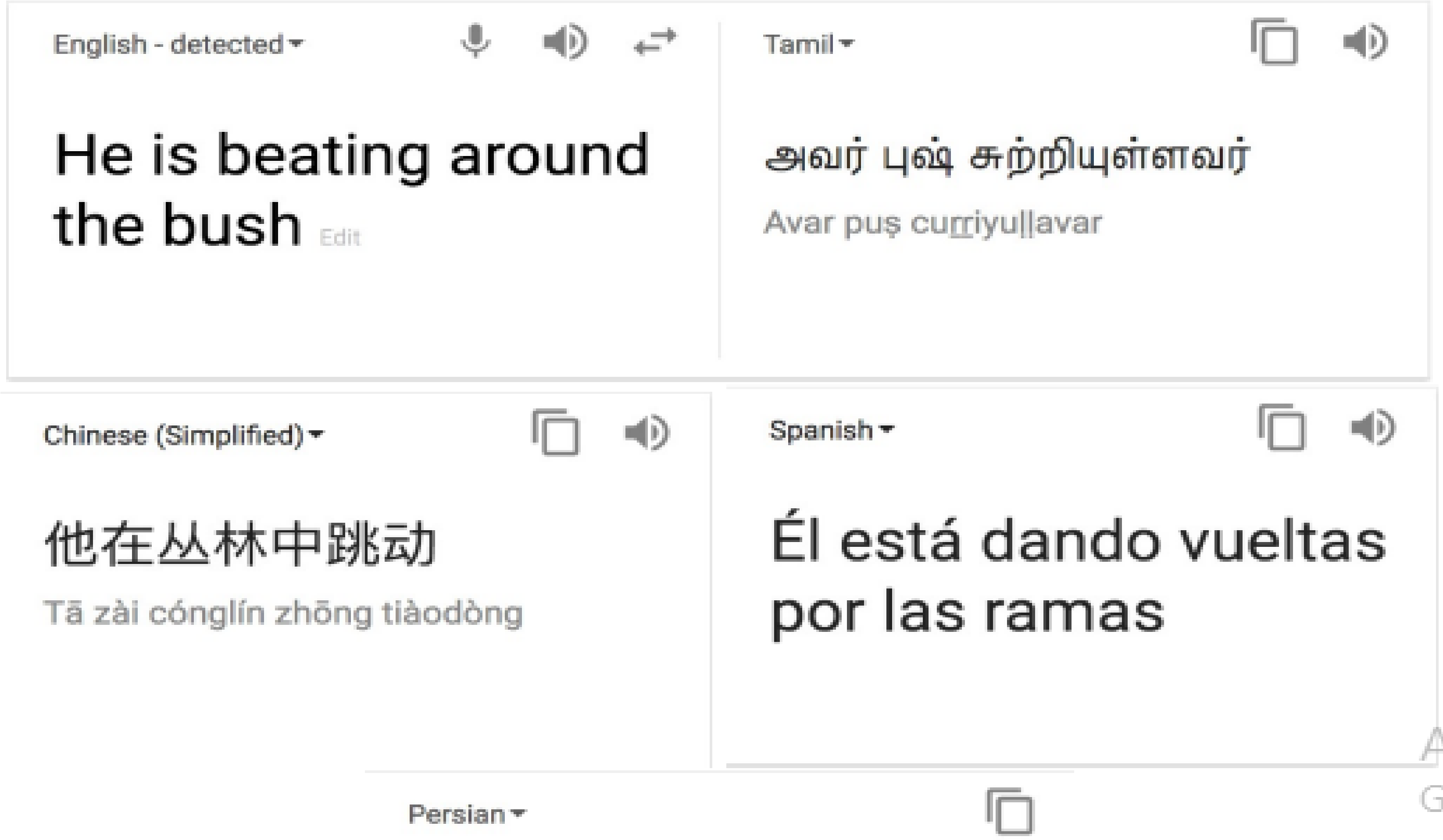
Problem at hand

Translate sentence s_1 from language l_1 to s_2 from language l_2

- l_1 : Source language
- l_2 : Target language
- s_1 : Sentence in source language
- s_2 : Sentence in target language
- ws_i : A Word in s_1
- wt_i : A Word in s_2

Challenges in Machine Translation

- Changes in the language structure
- Identifying compound words and two-word verbs
- Words with multiple meanings, Idioms and metaphors !!! This one is tricky !!



Solving the Machine Translation problem

Two problems to solve

- Given s_1, find best translations for each word in the target language
- Given these translations, find the right alignment

In most practical Machine Translation systems, we need to solve both the problems!

Classes of approaches

Classes of approaches

- Rule-based approach
- Corpus-based approach

Rule-based approach

- Uses manually curated rules for translation.
- Based on the morphological, syntactic and semantic map between the two languages.
- Laborious and time-consuming.

Corpus-based approach- Statistical Machine Translation

- Data driven methods that learn the rules of translation automatically from the corpus(data).
- A Statistical model is constructed who parameters are estimated from the corpus.
- Translations for unseen sentences are based on the learned model(s)

Statistical Machine Translation(SMT)

SMT model

$$\operatorname*{argmax}_{s_2 \in l_2} p(s_2|s_1) = \operatorname*{argmax}_{s_2 \in l_2} \underbrace{p(s_1|s_2)}_{\text{Translation model}} * \underbrace{p(s_2)}_{\text{Language model}}.$$

Translation model:

- $p(s_1|s_2)$ is the conditional prob. of the source sentence given the target sentence
- Likelihood of the source sentence s_1 given target s_2.
- Why are we modeling this? Think discriminative power of the model.

Language Model:

- $p(s_2)$ is the marginal prob. of the target sentence generated.(Think correctness of the sentence)
- Gives us how probable a given target sentence s_2 is, under the rules of the target language.

SMT - Translation model

- Models the relationship between source and target sentences.
- Harder part of the translation mechanism because of the number of unknowns as it models both translation and alignment.

How to estimate $p(s_1|s_2)$

- As discussed earlier, it's a 2 stage process.
- Stage 1: Given $ws_i \in s_1$ find the best $wt_j \in s_2$. This is word-by-word translation.
- Stage 2: Given the word-by-word translations, find the best alignment(or mapping) between $ws_i^{\prime}s$ and $wt_j^{\prime}s$.

Chicken and egg problem! We don't know the solution to either of the stages to begin with. EM!

Expectation Maximization(EM) for SMT

Stage1 - Expectation

Compute $p(a|s_1, s_2)$

Learning the best alignment function a that maps words ws_i to wt_j

Stage2 - Maximization

$wt_j = \text{argmax}_{wt \in l_2} p(w_t|ws_i)$

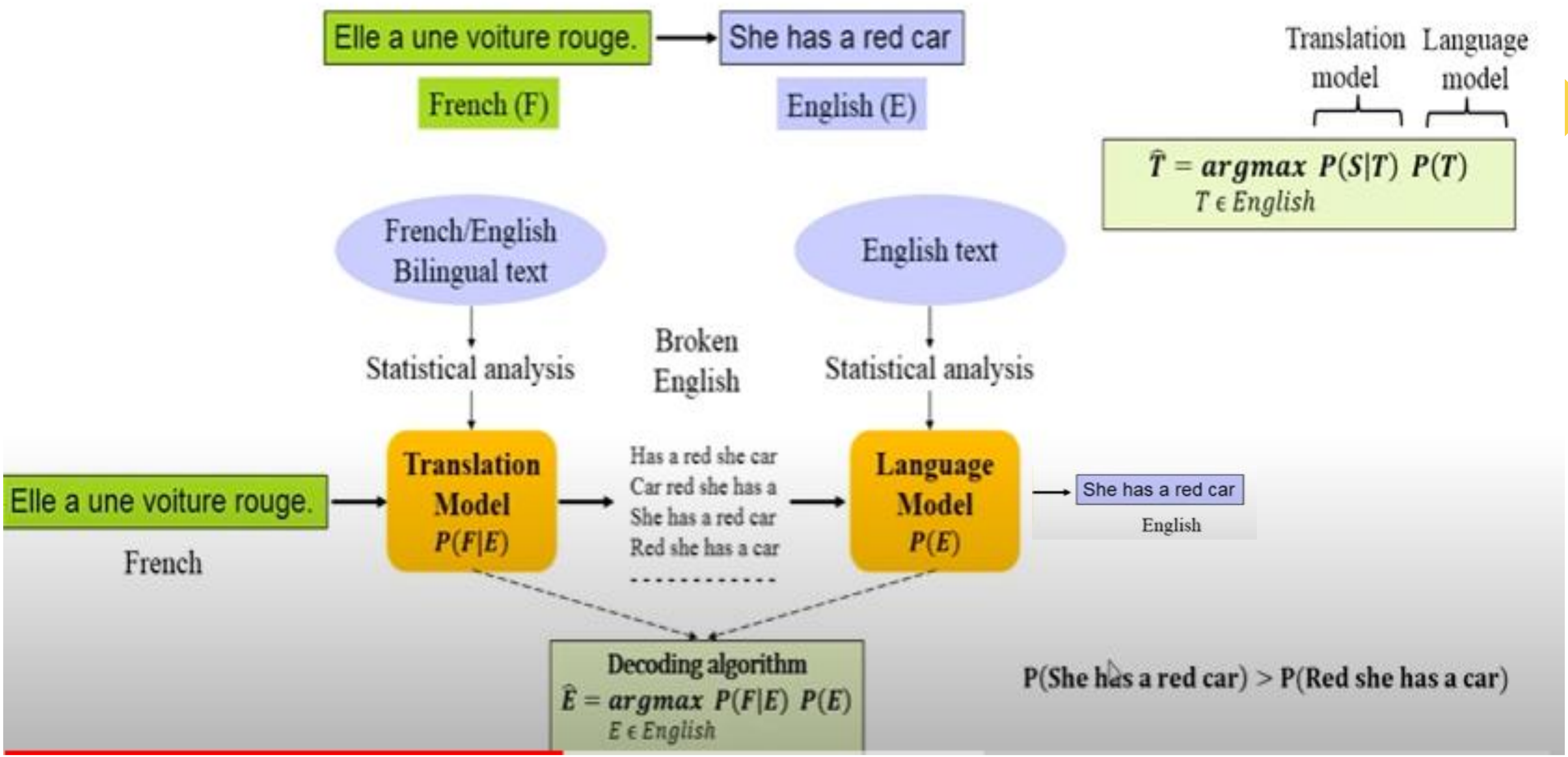
Given a training corpus and an alignment(mapping of word positions), estimating this is trivial(Maximization). Here it's just **MLE**.

- This falls into the framework of EM.
- The detailed algorithm is discussed later.
- Basic idea is to alternate between estimating optimal translations between words in the two sentences and finding the best possible alignment.

https://www.youtube.com/watch?v=LcnaxOnKSXI

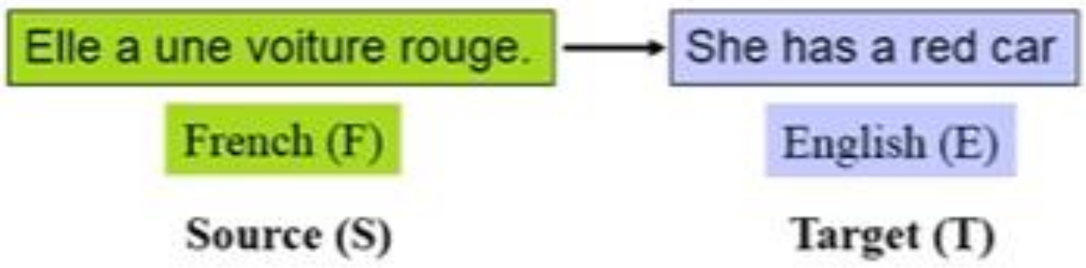
Approaches to MT

SMT



Approaches to MT

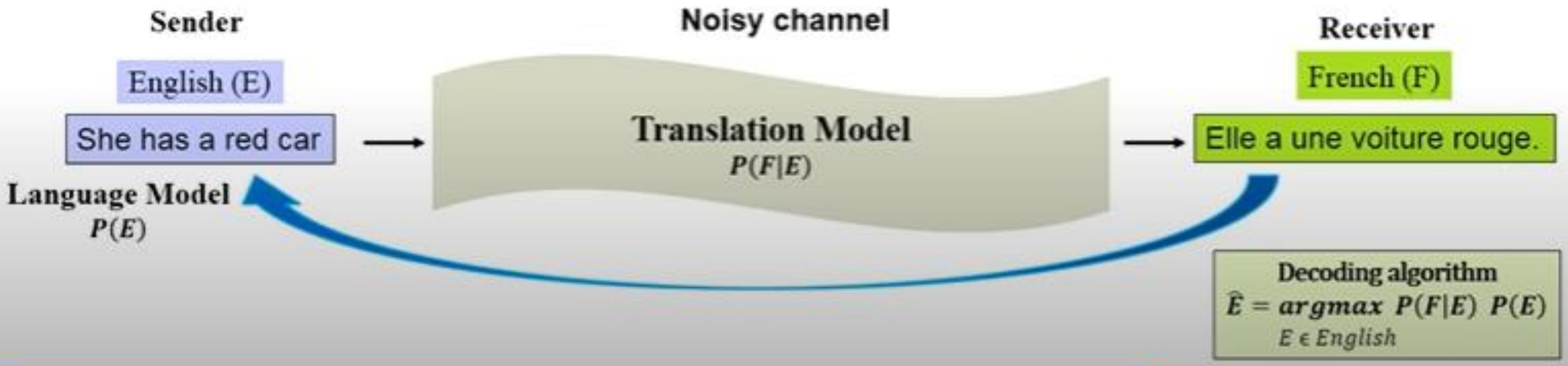
SMT



$$\hat{T} = \underset{T}{argmax}\, P(T|S)$$

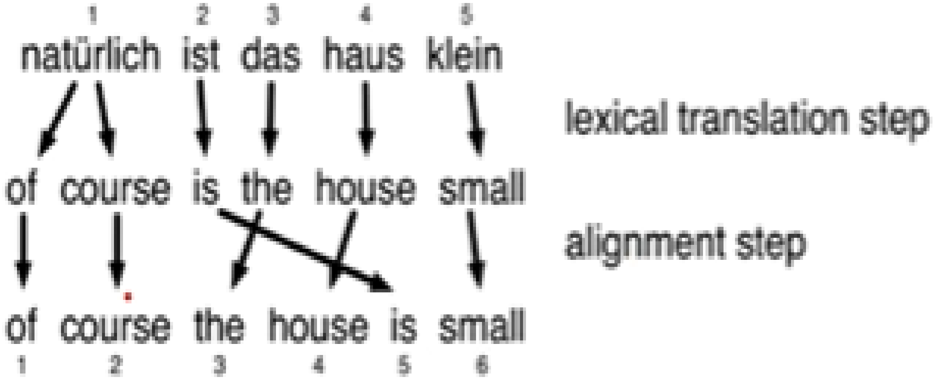
$$\hat{T} = \underset{T \in English}{argmax}\; P(S|T)\; P$$

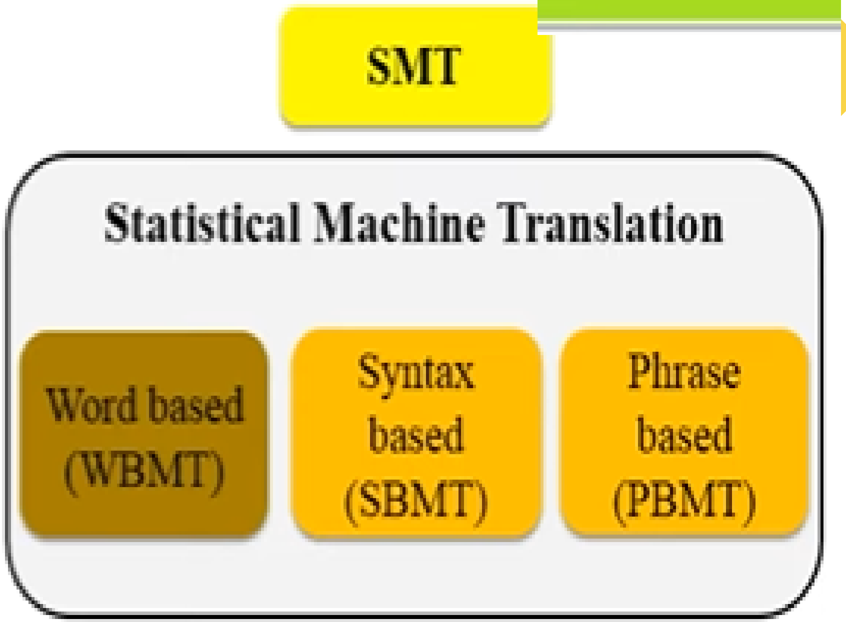
$$\hat{E} = \underset{E \in English}{argmax}\; P(F|E)\; P(E)$$



Approaches to MT: SMT (WBMT)

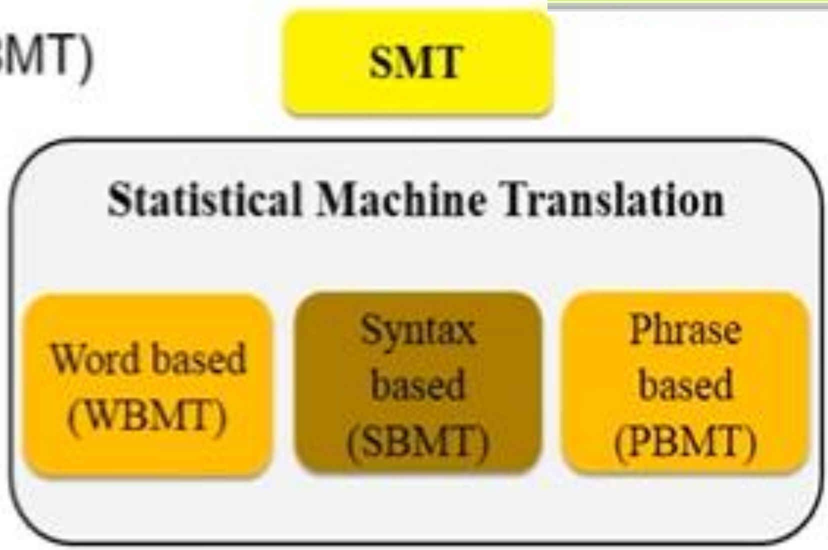
- Word is the unit of translation

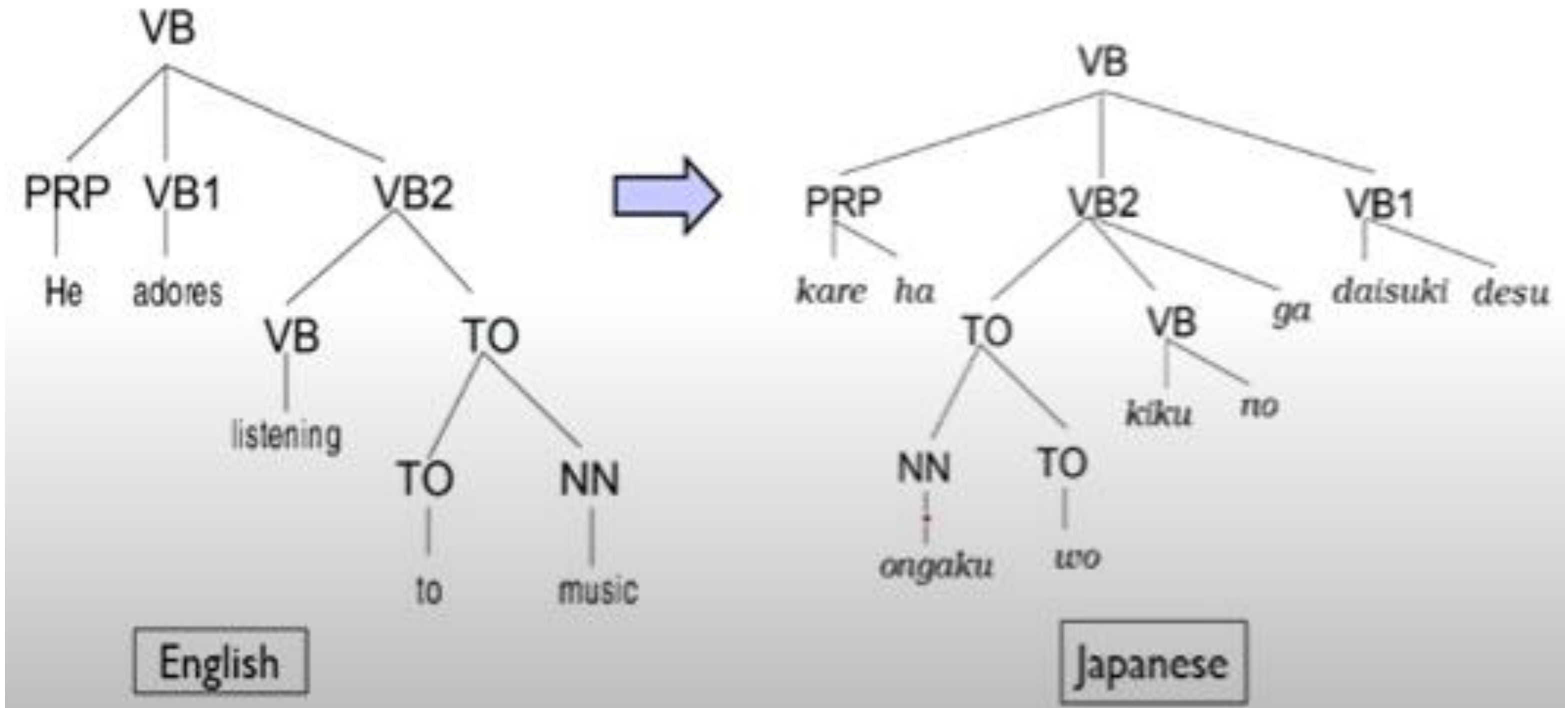




Approaches to MT: SMT (SBMT)

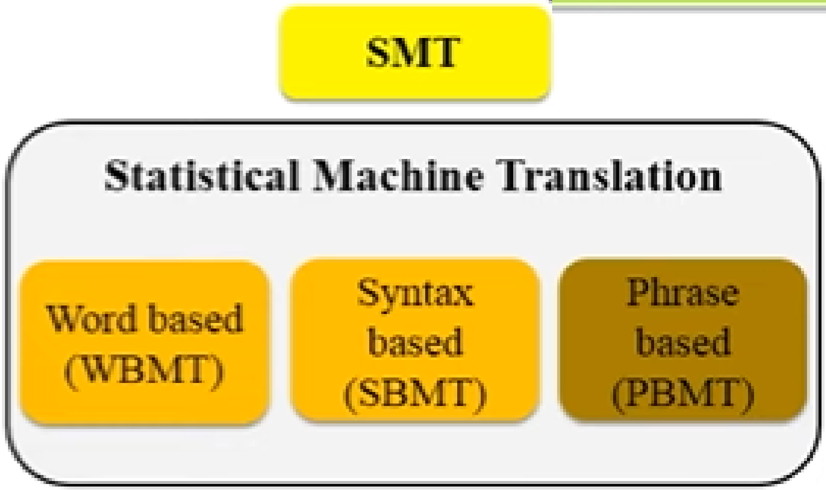
- Translating the syntactic units together, rather than single words (WBMT) or strings of words (PBMT)
  - i.e. parse trees of sentences/utterances.



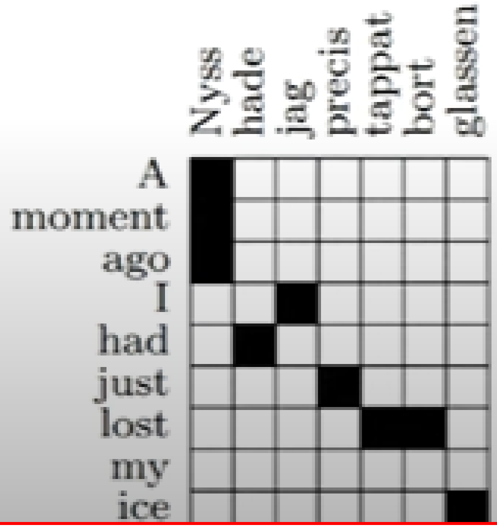


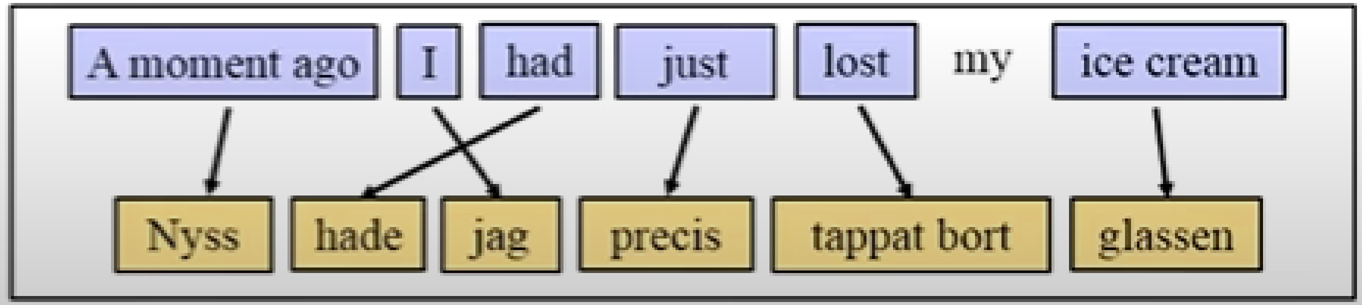
English (SVO) – Japanese (SOV)

Approaches to MT: SMT (PBMT)



- The intuition of phrase-based statistical MT is to use phrases (sequences of words) as well as single words as the fundamental units of translation.
- Steps:
  i. Phrase segmentation
  ii. Phrase translation
  iii. Output reordering





https://www.youtube.com/watch?v=My_aXkRIkfA

EM algorithm for SMT

Algorithm 1 EM Algorithm for SMT

Input: Set of sentence pairs (s_2, s_1)
Output: Translation prob. $p(wt_i|ws_j)$

initialize $p(wt|ws)$ uniformly
// initialize
while not converged **do**
  count$(wt_i|ws_j) = 0$ **for all** i, j
  total$(ws_j) = 0$ **for all** j
  for all sentence pairs (s_1, s_2) **do**
    // Compute normalization
    for all words wt_i in s_2 **do**
      s-total$(wt_i) = 0$
      for all words ws_i in s_1 **do**
        s-total$(wt_i) += p(wt_i|ws_j)$
      end for
    end for
  end for
  // estimate probabilities
  for all target words wt_i **do**
    for all source words ws_j **do**
      $p(wt_i|ws_j) = \frac{\text{count}(wt_i|ws_j)}{\text{total}(ws_i)}$
    end for
  end for
end while

Thank you